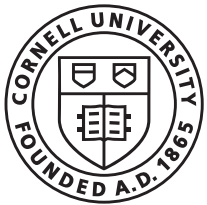


L4a: Graph and Tree Representations

CHEME 5800 Cornell University

Objectives for Today

Today we describe graph structure, calculate graph measures, and choose a storage form that matches the operation an algorithm performs.

Today's concepts

- Graph language and measures: define walks, paths, cycles, connectivity, degree, and density.
- Graph families: recognize complete graphs, bipartite graphs, and trees from their defining properties.
- Storage representations: compare edge lists, adjacency matrices, and adjacency lists by storage and access cost.

Lecture: Graph and Tree Representations

Lecture and Connected Exercises

Lecture: Graph and Tree Representations

Lab L4b: Breadth-First and Depth-First Search
Use an unweighted adjacency list to visit a directed graph with a queue or a stack.

Lecture L4c and lab L4d: Shortest-Path Algorithms
Retain edge weights and find minimum-cost routes through a process network.

Simple Graphs

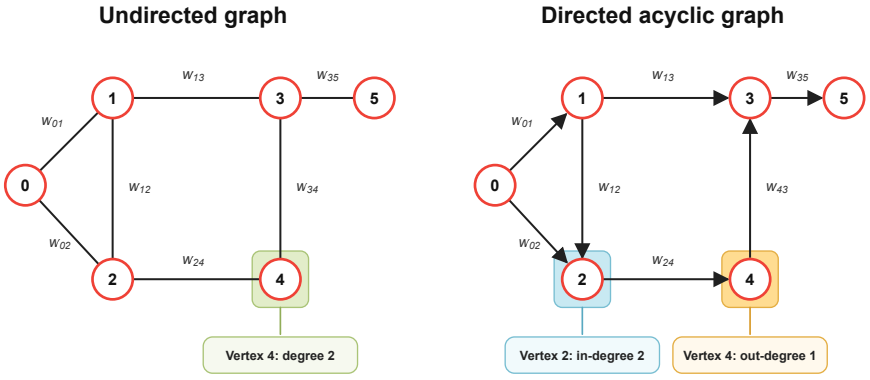
A simple graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a set of vertices $\mathcal{V}$ and a set of edges $\mathcal{E}$.

- In an undirected graph, an edge $\{u, v\}$ is an unordered pair of distinct vertices and may be followed either way.
- In a directed graph, an edge (u, v) is an ordered pair. It points from u to v, and (v, u) is a different edge.
- The word simple rules out self-loops and repeated parallel edges.
- An edge may carry a weight such as distance, capacity, time, or cost.

Modeling decision: direction records which transitions are feasible; weight records a quantity associated with an edge.

Direction Changes Which Routes Are Feasible

Six vertices and seven edges: |V| = 6, |E| = 7



The undirected drawing allows travel across an edge in either direction. The directed

Walks, Paths, and Cycles

Let $v_0, v_1, \ldots, v_k$ be a sequence of vertices.

- A walk follows an edge between every consecutive pair. A directed walk must follow every source-to-target direction.
- A path is a walk with no repeated vertices.
- A cycle is a closed walk with at least one edge that repeats no edge and no vertex except $v_0 = v_k$.

A path never revisits a vertex. A cycle returns to its starting vertex.

Connectivity Depends on Direction

Connectivity asks whether paths join the vertices.

- An undirected graph is connected when a path joins every pair of vertices.
- A directed graph is weakly connected when ignoring edge directions produces a connected undirected graph.
- A directed graph is strongly connected when every ordered pair (u, v) is joined by a directed path from u to v.

In the figure, vertex 0 can reach vertex 5, but vertex 5 has no outgoing edge. The directed graph is weakly but not strongly connected. It has no directed cycles, so it is a directed acyclic graph.

Directed Degree Has Two Parts

Count the edges touching vertex 4. Only its incident edges are shown.

Undirected

Two edges touch vertex 4.

$$\deg(v_4) = 2$$

Directed

One edge enters; one leaves.

$$\deg^{\text{in}}(v_4) = 1, \qquad \deg^{\text{out}}(v_4) = 1$$

Total degree: $\deg(v_4) = \deg^{\text{in}}(v_4) + \deg^{\text{out}}(v_4) = 2.$

Degree Counts Recover the Edge Count

Let $n = |\mathcal{V}|$ and $m = |\mathcal{E}|$. In an undirected graph, summing degrees counts both endpoints of every edge:

$$\underbrace{\sum_{v_i \in \mathcal{V}} \deg(v_i)}_{\text{all incident ends}} = 2m.$$

In a directed graph, each edge contributes once to each directional sum:

$$\underbrace{\sum_{v_i \in \mathcal{V}} \deg^{\text{in}}(v_i)}_{\text{all targets}} = \underbrace{\sum_{v_i \in \mathcal{V}} \deg^{\text{out}}(v_i)}_{\text{all sources}} = m.$$

These handshaking identities are useful consistency checks for graph data.

Average Degree Lies Between the Extremes

For an undirected graph with $n \geq 1$, let $\delta(\mathcal{G})$ and $\Delta(\mathcal{G})$ be its minimum and maximum degrees. The average degree is given by:

$$\underbrace{\bar{d}(\mathcal{G}) = \frac{1}{n} \sum_{v_i \in \mathcal{V}} \deg(v_i) = \frac{2m}{n}}_{\text{average degree}}.$$

It is bounded by:

$$\delta(\mathcal{G}) \leq \bar{d}(\mathcal{G}) \leq \Delta(\mathcal{G}).$$

An undirected graph is regular of degree r when every vertex has degree r. Then $\delta = \bar{d} = \Delta = r$.

Density Measures the Fraction of Possible Edges

A simple loop-free graph on $n \geq 2$ vertices has maximum edge count:

$$|\mathcal{E}|_{\max} = \begin{cases} n(n-1)/2, & \text{undirected,} \\ n(n-1), & \text{directed.} \end{cases}$$

The density is the observed fraction of that maximum:

$$\underbrace{\rho(\mathcal{G}) = \frac{|\mathcal{E}|}{|\mathcal{E}|_{\max}}}_{\text{graph density}}, \qquad 0 \leq \rho(\mathcal{G}) \leq 1.$$

Density near one is dense; density near zero is sparse. This lecture sets $\rho(\mathcal{G}) = 0$ when $n < 2$.

Four Measures Describe Graph Arrangement

For an undirected graph, counts can be supplemented by:

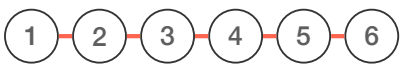
- Diameter $\mathrm{diam}(\mathcal{G})$: the largest shortest-path distance in a connected graph, measured in edges.
- Clique number $\omega(\mathcal{G})$: the largest number of vertices that are all pairwise adjacent.
- Chromatic number $\chi(\mathcal{G})$: the fewest colors that place different colors on adjacent vertices.
- Independence number $\alpha(\mathcal{G})$: the largest number of vertices with no edge between any selected pair.

Computational consequence: these measures require edge lookups and neighbor iteration; vertex and edge counts alone are insufficient.

Path and Star Graphs

Both graphs have $n = 6$ vertices and $n - 1 = 5$ edges.

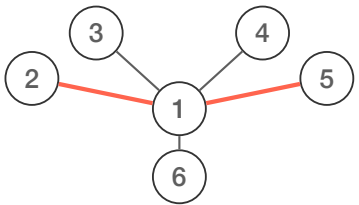
Path graph



Maximum degree: $\Delta = 2$

Diameter: $n - 1 = 5$ edges

Star graph



Maximum degree: $\Delta = n - 1 = 5$

Diameter: 2 edges

The highlighted route joins a most distant pair of vertices.

The graph is a mathematical object. Its representation determines what an algorithm can access efficiently.

Three Forms Store the Same Graph

Let $n = |\mathcal{V}|$ and $m = |\mathcal{E}|$.

- An edge list stores one source-target-weight record per edge.
- An adjacency matrix stores one entry for every ordered pair of vertices.
- An adjacency list stores one outgoing-neighbor collection per vertex.

Each form can describe the same graph, provided its vertex set is retained. The representation affects edge lookup and neighbor iteration.

Edge Lists Store One Record per Edge

An edge list stores each edge's endpoints and, when present, its weight.

- Edge records require $\Theta(m)$ storage. An explicit vertex set adds $\Theta(n)$, giving $\Theta(n + m)$ overall.
- Isolated vertices do not appear in edge records and must be stored separately.
- Without an index, finding an edge or collecting a vertex's neighbors takes $\Theta(1 + m)$ time in the worst case.

Edge lists suit reading, writing, and scanning the complete edge collection.

Adjacency Matrices Reserve Every Vertex Pair

For vertex order $v_1, \ldots, v_n$, an unweighted adjacency matrix has entries:

$$a_{ij} = \begin{cases} 1, & \text{an edge joins } v_i \text{ to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

In a weighted matrix, an existing edge's entry stores its weight w_{ij}.

- Storage is $\Theta(n^2)$ and one edge lookup is $\Theta(1)$.
- Listing neighbors of v_i scans row i in $\Theta(n)$ time.
- An undirected matrix is symmetric; a directed matrix need not be.

Zero weights: zero can mark a missing edge only when it is not a valid edge weight; otherwise use a separate marker.

Adjacency Lists Store Neighbors

An adjacency list stores one collection of outgoing neighbors for each source.

- Unweighted form: `source => [target, ...]`.
- Weighted form: `source => [(target, weight), ...]`.
- Storage is $\Theta(n + m)$ for a directed graph.
- Accessing a source's dictionary entry is expected $\Theta(1)$; iterating over its out-neighbors costs $\Theta(1 + \deg^{\text{out}}(v_i))$.

In an undirected list, each edge appears in both endpoint lists. The exact count is $n + 2m$ vertex and neighbor records, which remains $\Theta(n + m)$.

Pair Lists with an Edge-Weight Map

A flat dictionary can map a known endpoint pair directly to a weight:

$$(u, v) \longmapsto w(u, v).$$

- Expected lookup for a known (u, v) is $\Theta(1)$.
- Finding all neighbors of u still requires scanning m entries.
- Pairing the map with an unweighted adjacency list gives fast direct weight lookup and fast neighbor iteration.

Design principle: one graph object may maintain more than one representation when its algorithms require different access patterns.

Storage and Access Costs

Let $d_u = \deg^{\text{out}}(u)$. For a directed graph:

Representation	Storage	Find (u, v)	List neighbors
Edge list + vertex set	$\Theta(n + m)$	$\Theta(1 + m)$	$\Theta(1 + m)$
Adjacency matrix	$\Theta(n^2)$	$\Theta(1)$	$\Theta(n)$
Adjacency list	$\Theta(n + m)$	$\Theta(1 + d_u)$	$\Theta(1 + d_u)$

Edge searches use worst-case scan lengths; dictionary access is expected constant time. The added 1 includes the work needed for an empty list.

Weight lookup: a separate edge-weight map provides expected $\Theta(1)$ access to a known vertex pair.

Choose the Representation from the Operation

Does the edge (u, v) exist? Adjacency matrix
Read A_{uv} directly in $\Theta(1)$ time. Repeated edge lookups favor a matrix when its $\Theta(n^2)$ storage is affordable, especially for dense graphs.

Which vertices are neighbors of u? Adjacency list
Scan the d_u neighbors of u instead of all n matrix entries. For a sparse graph, this reduces neighbor-search work and uses $\Theta(n + m)$ storage.

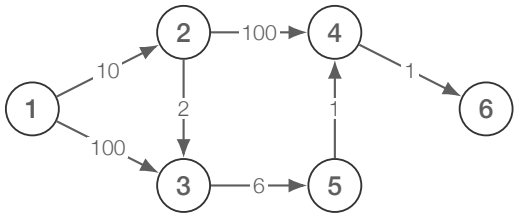
How do we process every edge? Edge list
Read, write, or sort the edges as records. A full scan takes $\Theta(m)$ time, but finding one vertex's neighbors also requires scanning the list.

Balance the work done most often against the storage available.

Worked Example: A Six-Vertex Directed Graph

The file `SimpleGraph.txt` contains seven weighted edge records:

source	target	weight
1	2	10
1	3	100
2	3	2
2	4	100
3	5	6
4	6	1
5	4	1



Notebook functions: `read_weighted_edges`, `adjacency_list`, and `adjacency_matrix`.

The Same Edges Become a List or a Matrix

Outgoing adjacency list

$1 \longrightarrow [2, 3]$
$2 \longrightarrow [3, 4]$
$3 \longrightarrow [5]$
$4 \longrightarrow [6]$
$5 \longrightarrow [4]$
$6 \longrightarrow [\,]$

This lecture's helper stores targets only; the original edge records retain the weights.

Weighted adjacency matrix

	1	2	3	4	5	6
1	0	10	100	0	0	0
2	0	0	2	100	0	0
3	0	0	0	0	6	0
4	0	0	0	0	0	1
5	0	0	0	1	0	0
6	0	0	0	0	0	0

Vertex Labels and Matrix Indices

Relabel the same six vertices as $10, 20, 30, 40, 50, 60$; keep all edges and weights. Choose `vertex_order` $= [10, 20, 30, 40, 50, 60]$.

Index-to-vertex mapping

Index i	Vertex label
1	10
2	20
3	30
4	40
5	50
6	60

Weighted adjacency matrix

$$A = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 10 & 100 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 100 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 6 & 0 \\ 4 & 0 & 0 & 0 & 0 & 0 & 1 \\ 5 & 0 & 0 & 0 & 1 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$A_{1,3} = 100$ records the edge $10 \to 30$ with weight 100.

Graph Storage: Counts and Growth

Our directed example has $n = 6$, $m = 7$, and $\rho = 7/[6(6-1)] = 7/30$.

Graph	Adjacency matrix	Adjacency list
Six-vertex example	$n^2 = 36$ entries	$n + m = 13$ items
Sparse: $m = 10n$	$\Theta(n^2)$	$\Theta(n + 10n) = \Theta(n)$
Dense: $m = \Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$

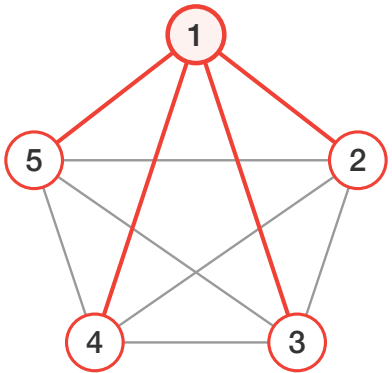
The matrix reserves space for absent edges. The example's list stores 6 vertex collections and 7 neighbor entries.

Big Theta describes storage growth as well as running time. With 10 outgoing edges per vertex, doubling n approximately doubles list storage but quadruples matrix storage.

Three graph families turn structural definitions into recognition tests, edge counts, and storage requirements.

Complete Graphs Use Every Possible Edge

A complete graph K_n is a simple undirected graph with an edge between every pair of distinct vertices.



What the picture shows

Degree of every vertex

$$\deg(v) = 5 - 1 = 4$$

Number of edges

$$|\mathcal{E}| = \frac{5(4)}{2} = 10$$

Fraction of possible edges

$$\rho(K_5) = 1$$

edges incident to vertex 1

Complete-Graph Measures Follow Directly

For $n \geq 2$, every possible edge is present:

$$|\mathcal{E}| = \binom{n}{2} = \frac{n(n-1)}{2}, \qquad \deg(v_i) = n - 1.$$

Consequently:

$$\rho(K_n) = 1, \qquad \operatorname{diam}(K_n) = 1,$$

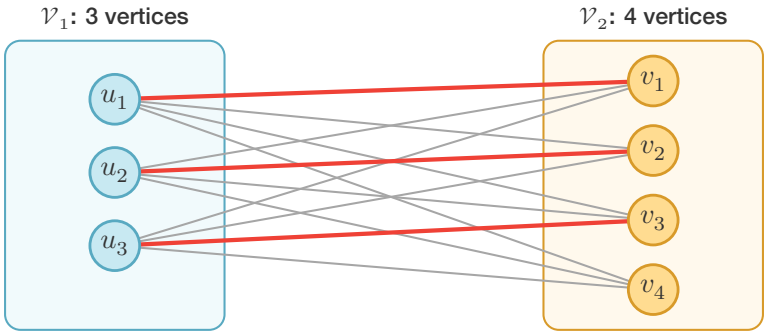
$$\omega(K_n) = \chi(K_n) = n, \qquad \alpha(K_n) = 1.$$

Round-robin schedules have the edge pattern of K_n. Traveling-salesman models use a weighted K_n when every city pair has a symmetric travel cost.

Storage: since $|\mathcal{E}| = \Theta(n^2)$, both matrices and adjacency lists require $\Theta(n^2)$ storage.

Bipartite Graphs Split Vertices into Two Groups

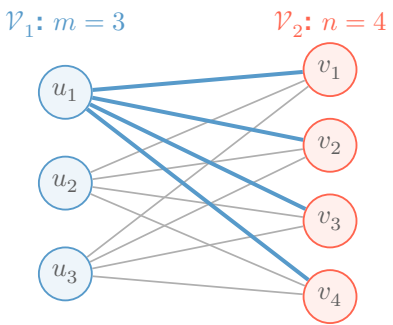
A graph is bipartite when $\mathcal{V} = \mathcal{V}_1 \dot{\cup} \mathcal{V}_2$ and every edge has one endpoint in each group.



Complete Bipartite Graphs

A complete bipartite graph $K_{m,n}$ includes every edge between its two groups and no edge within a group.

The graph $K_{3,4}$



Count each edge from the left

Each of m left vertices connects to all n right vertices:

$$|\mathcal{E}| = mn = 3(4) = 12.$$

Degree counts the opposite group

$$\deg(v) = \begin{cases} n = 4, & v \in \mathcal{V}_1, \\ m = 3, & v \in \mathcal{V}_2. \end{cases}$$

The four blue edges meet u_1.

For $m, n \geq 1$, $K_{m,n}$ is regular exactly when $m = n$; $K_{3,4}$ is not regular.

Three Equivalent Tests Identify Bipartite Graphs

For an undirected graph $\mathcal{G}$, the following statements are equivalent:

1. $\mathcal{G}$ is bipartite.
2. $\mathcal{G}$ can be colored with at most two colors, so $\chi(\mathcal{G}) \leq 2$.
3. $\mathcal{G}$ contains no odd-length cycle.

Start a traversal in each uncolored component. Give each uncolored neighbor the opposite color and reject any same-color edge. With an adjacency list, the test takes $\Theta(|\mathcal{V}| + |\mathcal{E}|)$ time.

Matchings Assign Distinct Options

A matching is a set of edges that share no vertices. A matching that covers $\mathcal{V}_1$ assigns a different right-side option to every left-side vertex.

For $S \subseteq \mathcal{V}_1$, let $N(S) \subseteq \mathcal{V}_2$ be all neighbors of vertices in S. A matching covering $\mathcal{V}_1$ exists if and only if:

$$|N(S)| \geq |S| \qquad \text{for every } S \subseteq \mathcal{V}_1.$$

Every group on the left must collectively have at least as many available options on the right. A perfect matching covers both groups and therefore requires $|\mathcal{V}_1| = |\mathcal{V}_2|$ together with Hall's condition.

Hall's theorem: Mathematics for Computer Science, MIT

Bipartite Matrices Have Block Structure

Order the m vertices of $\mathcal{V}_1$ before the n vertices of $\mathcal{V}_2$. For a directed bipartite graph:

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{C} & \mathbf{0} \end{bmatrix}, \qquad \mathbf{B} \in \mathbb{R}^{m \times n}, \quad \mathbf{C} \in \mathbb{R}^{n \times m}.$$

- $\mathbf{B}$ records edges from $\mathcal{V}_1$ to $\mathcal{V}_2$; $\mathbf{C}$ records the reverse direction.
- The diagonal blocks are zero: no edge stays within either group.
- If all edges point from $\mathcal{V}_1$ to $\mathcal{V}_2$, then $\mathbf{C} = \mathbf{0}$.
- For an undirected graph, $\mathbf{C} = \mathbf{B}^\mathsf{T}$.

Biadjacency matrix: for undirected $K_{3,4}$, $\mathbf{B}$ is a 3×4 matrix of ones; store $\mathbf{B}$ and the vertex order.

Trees Are Minimally Connected Graphs

A tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ is a connected undirected graph with no cycles.

- Exactly one path joins every pair of vertices.
- Removing any edge disconnects the tree.
- Adding an edge between nonadjacent vertices creates exactly one cycle.

A tree has exactly enough edges to remain connected and no edge that can be removed without losing connectivity.

Tree Tests, Density, and Storage

For a simple undirected graph with $n \geq 1$:

Six equivalent tests

1. The graph is a tree.
2. It is connected and has $n - 1$ edges.
3. It has no cycle and has $n - 1$ edges.
4. It is connected, and removing any edge disconnects it.
5. It has no cycle, and adding any missing edge creates one cycle.
6. Every pair of vertices is joined by exactly one path.

Density and storage

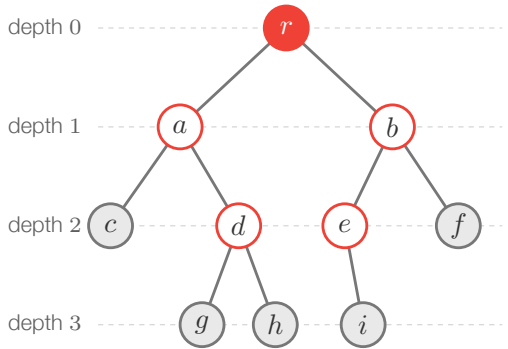
Every tree has $m = n - 1$ edges. For $n \geq 2$:

$$\rho(\mathcal{T}) = \frac{n-1}{n(n-1)/2} = \frac{2}{n}.$$

As $n \to \infty$, $\rho(\mathcal{T}) \to 0$: the tree stays connected while becoming sparse.

Adjacency-list storage is $\Theta(n)$ because $m = n - 1$.

A Root Gives a Tree Parent-Child Structure



Legend

- **Root** r — has no parent; depth is 0
- **Every other vertex** — has exactly one parent
- **Leaves** c, f, g, h, i — have no children
- **Height** $= 3$ — the largest vertex depth

Every non-root vertex has one parent. Vertices may have children; those with no children are leaves. Depth counts edges from the root, and height is the maximum depth.

Spanning Trees Retain Connectivity

Every connected undirected graph contains a spanning tree: a subgraph with all its vertices that remains connected and has no cycles.

- A spanning tree on n vertices contains exactly $n - 1$ edges.
- A forest is an undirected graph with no cycles.
- Each connected component of a forest is a tree.

Summary

We described graph structure, examined three graph families, and compared representations for storing their connections.

Key takeaways

- **Graph structure and measures.** We used connectivity, degree, and density to describe graphs, then compared arrangements with the same vertex and edge counts.
- **Graph families.** We related complete graphs, bipartite graphs, and trees to edge counts, coloring, matchings, and unique paths.
- **Storage and access.** We compared edge lists, matrices, and adjacency lists to explain their memory use and access costs.

L4b turns an adjacency list into a systematic visit order; L4c uses weights and minimizes route cost.

That's a wrap. Which graph representation would you choose for the next algorithm, and what operation justifies that choice?